Blood flukes (Schistosomes)

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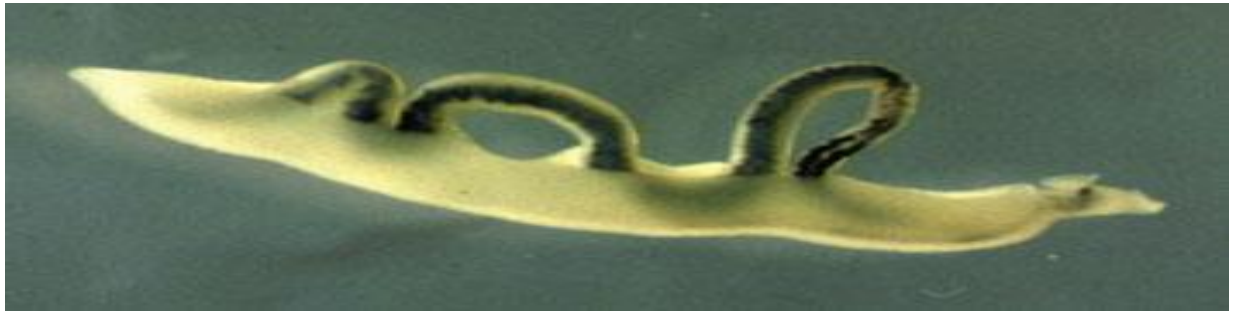
Third Stage

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BLOOD FLUKES (**Schistosomes)**

- Schistosomes are dioecious, (sexes are separate) trematodes, which lead to Schistosomiasis (bilharziasis).
- Schistosomes were formerly called Bilharzia after Theodor Bilharz who in 1851, first observed the worm in the mesenteric veins of an Egyptian in Cairo.
- It is estimated that over 100 million people are infected with *Schistosoma haematobium*, *Schistosoma mansoni*, and *Schistosoma japonicum.*
- Annually more than 200,000 deaths occur.

- Schistosomiasis is a water-borne disease constituting an important public health problem and affecting millions of persons in Africa, Asia, and Latin America.
- The male worm is broader than the female and its lateral borders are rolled ventrally into a cylindrical shape, producing a long groove or trough called the gynecophoric canal, in which the female is held.



- It appears as though the body of the male is split longitudinally to produce this canal; hence the name schistosome (Greek schisto: split and soma: body)

Main types of Schistosomes

- *Schistosoma haematobium*
- *Schistosoma mansoni*
- *Schistosoma japonicum.*
- *Schistosoma mekongi*
- *Schistosoma intercalatum.*

- All schistosomes live in venous plexuses in the body of the definitive host, the location varying with the species.

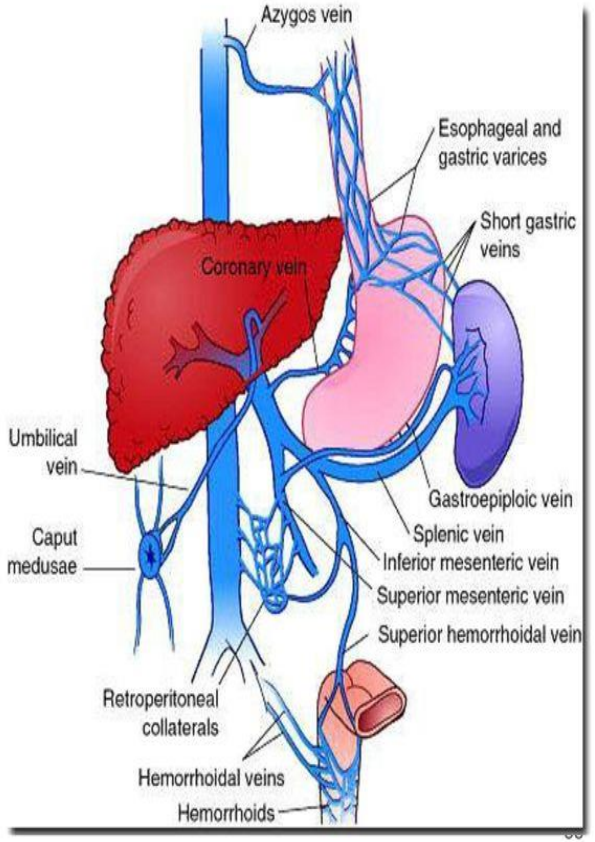
- ***Schistosoma mansoni***
- Adult worm lives in the inferior mesenteric vein (ileocecal region).
- ***Schistosoma japonicum***
- Adult worms are seen typically in the venules of the superior mesenteric vein draining the ileocecal region. they are also seen in the intrahepatic portal venules and hemorrhoidal plexus of veins.

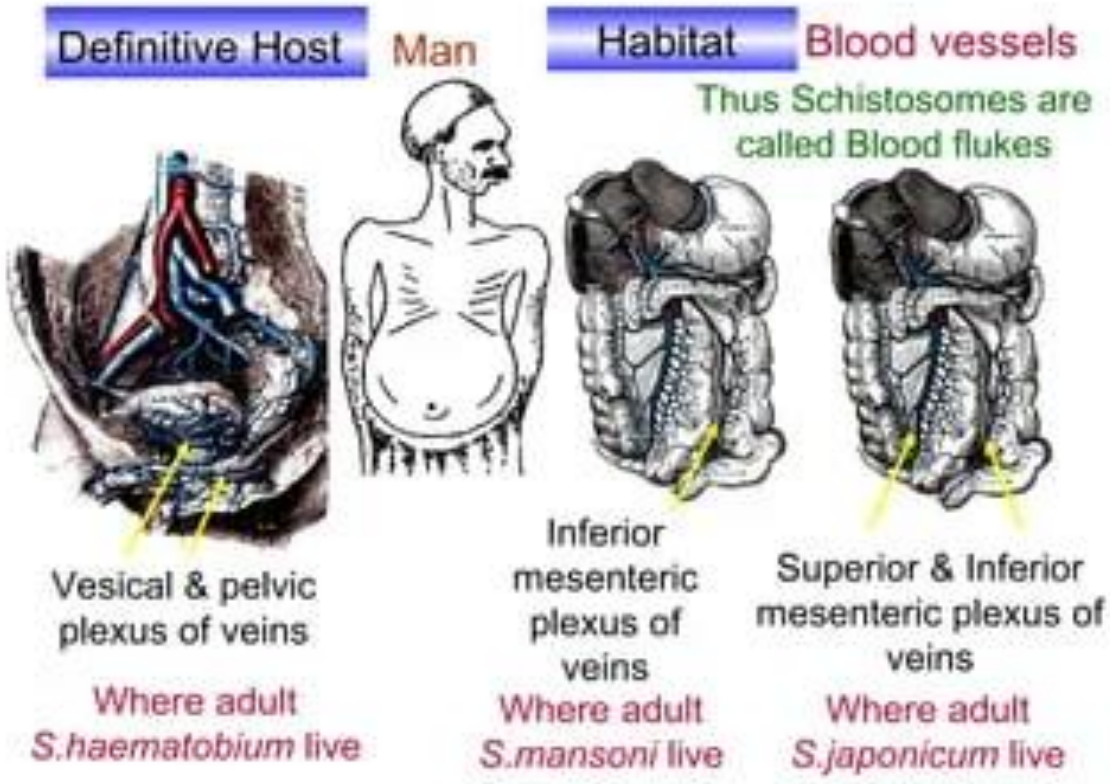
} **intestinal schistosomiasis**

- ***Schistosoma haematobium*** → **urinary schistosomiasis**
- The adult worms live in the vesical and pelvic plexuses of veins draining the urinary bladder.

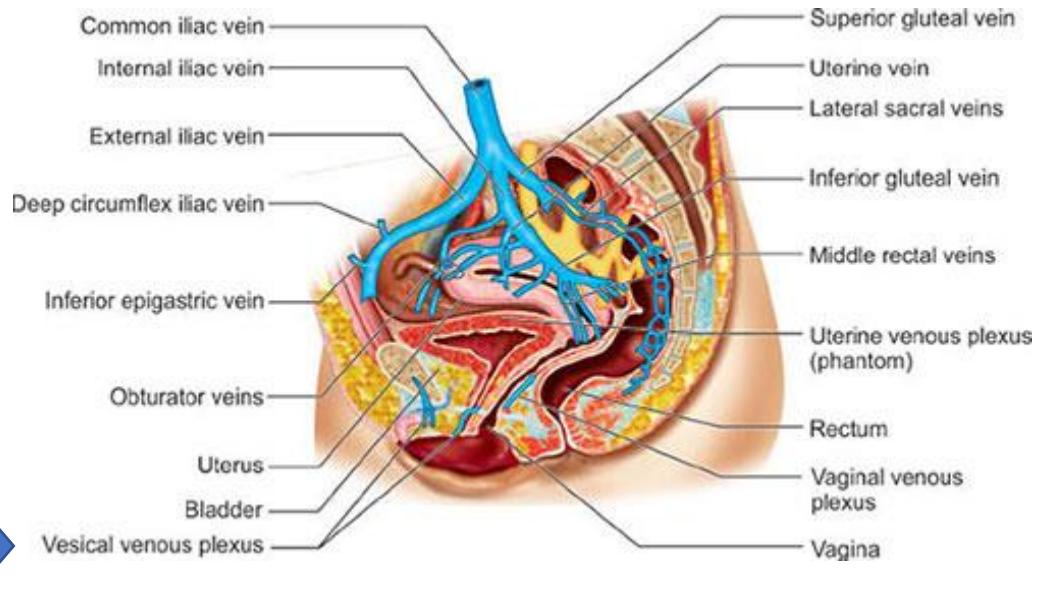
Habitat

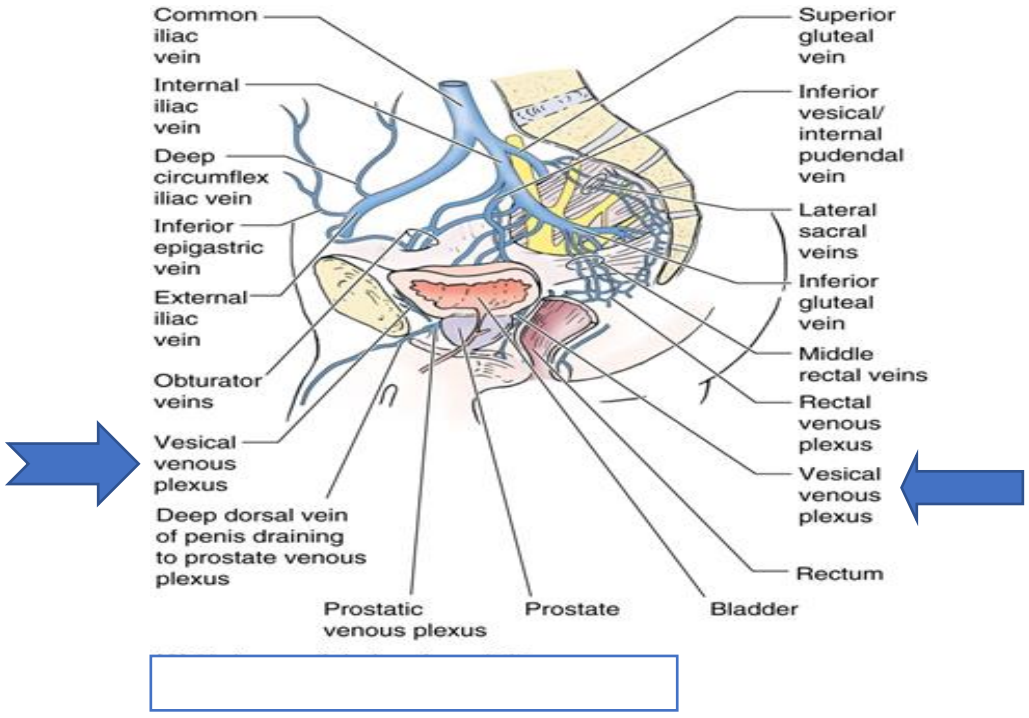
- Hepatic fibrosis is **caused by** a granulomatous reaction to Schistosoma **eggs** that have been carried to the liver.

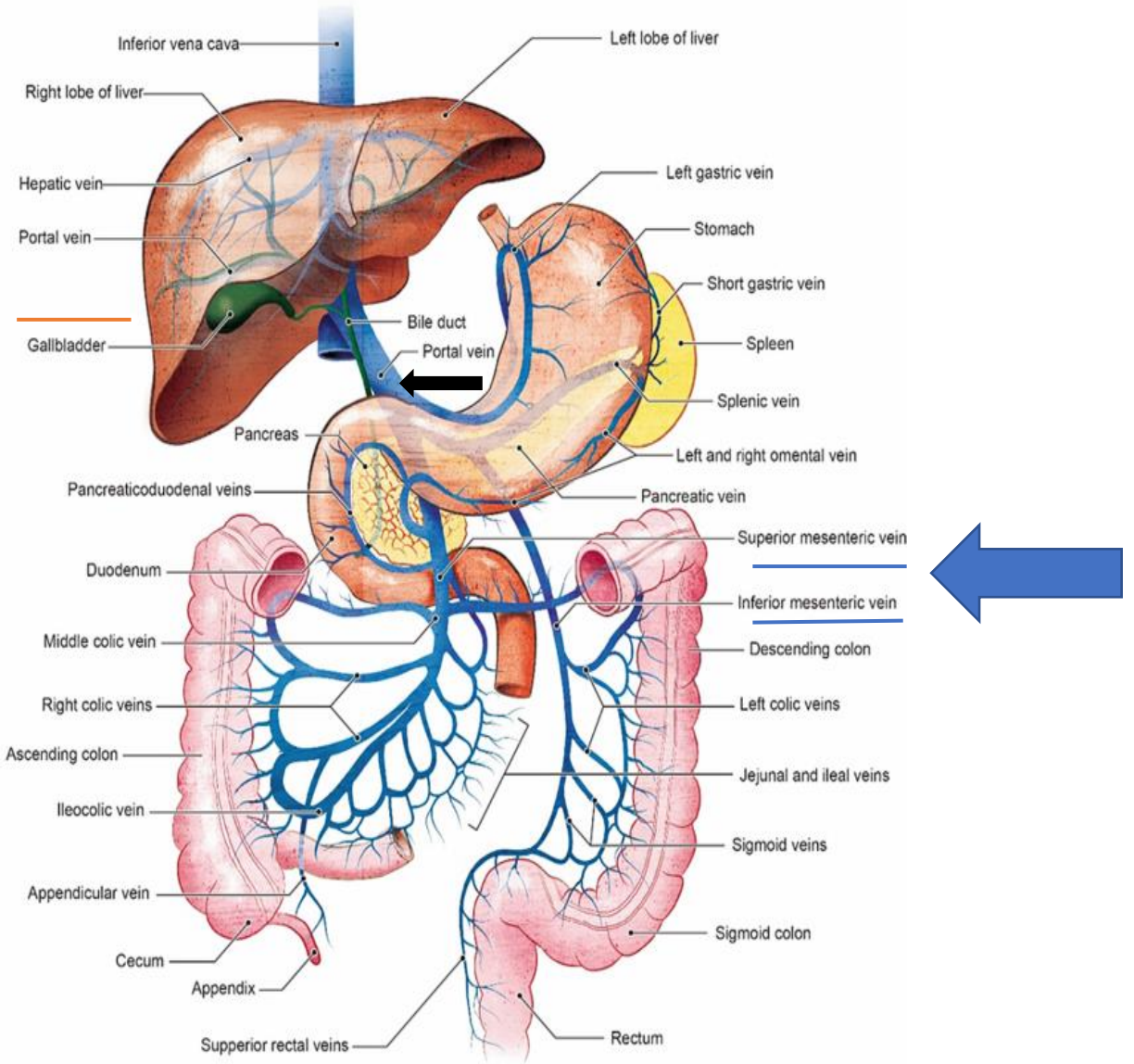




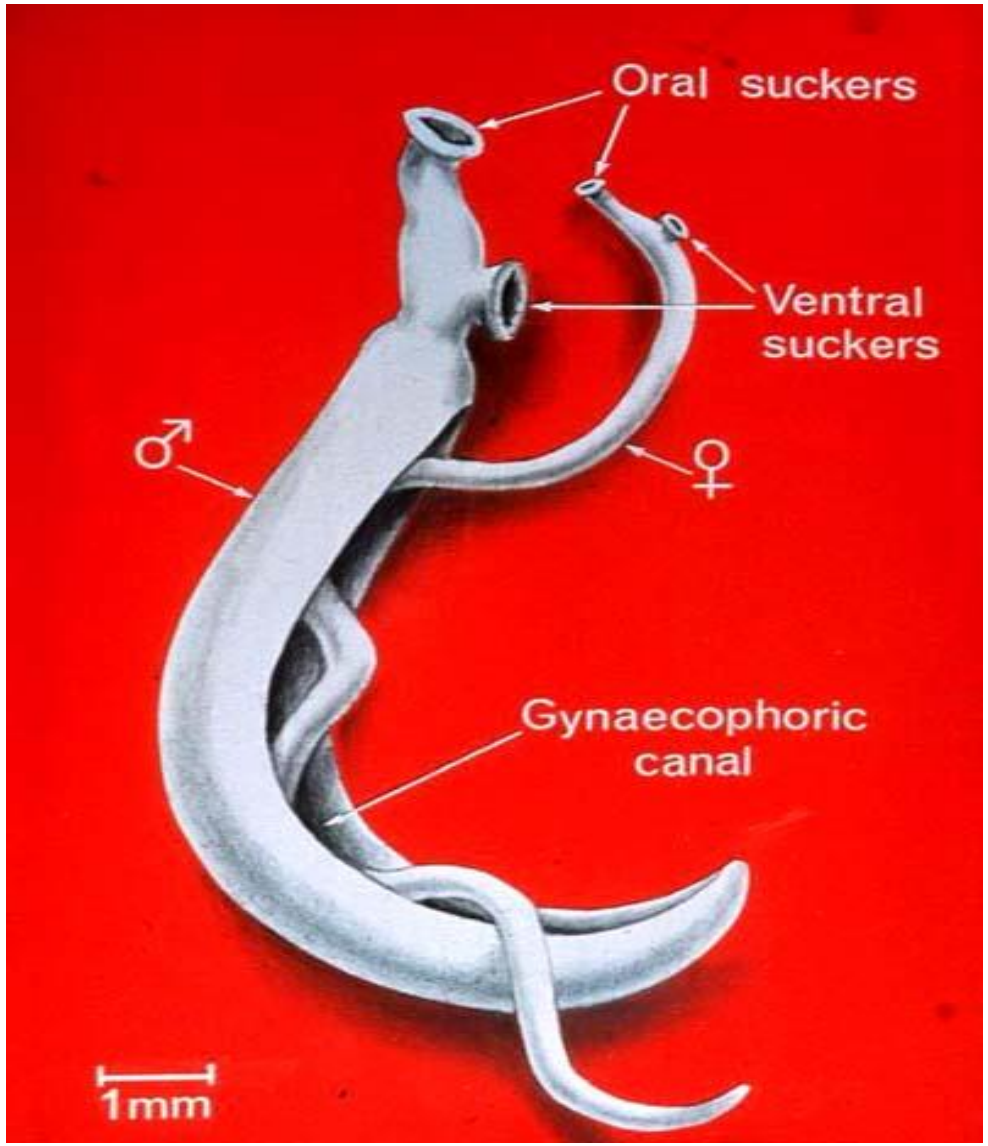
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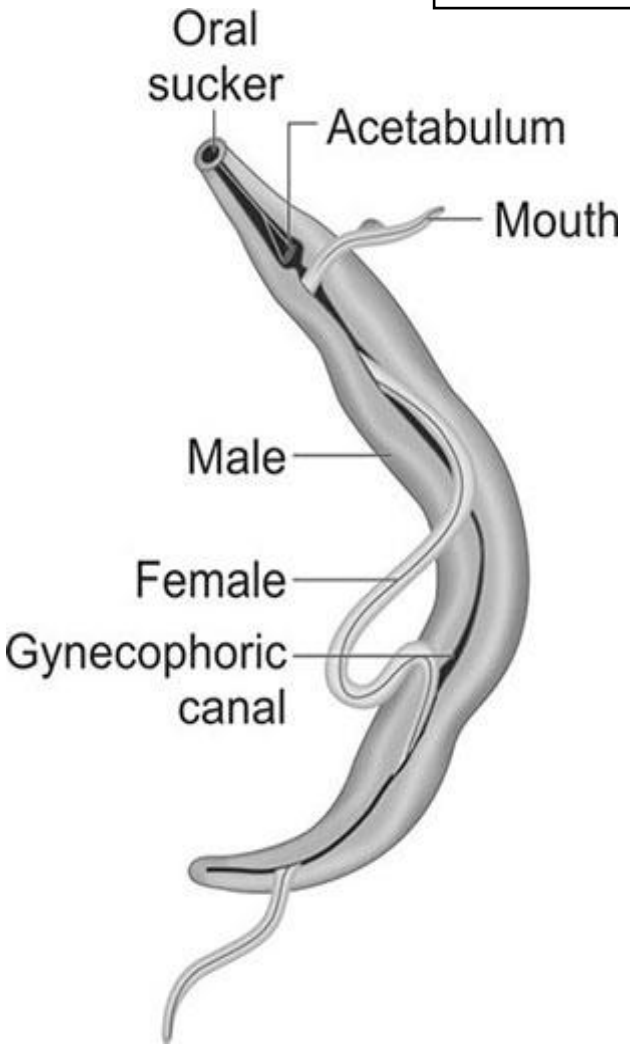
Schistosomes adult worm





The male is 10–15 mm
The female is longer 17-20 mm

Morphology of Adult *Schistosoma*



Schistosoma is _____.

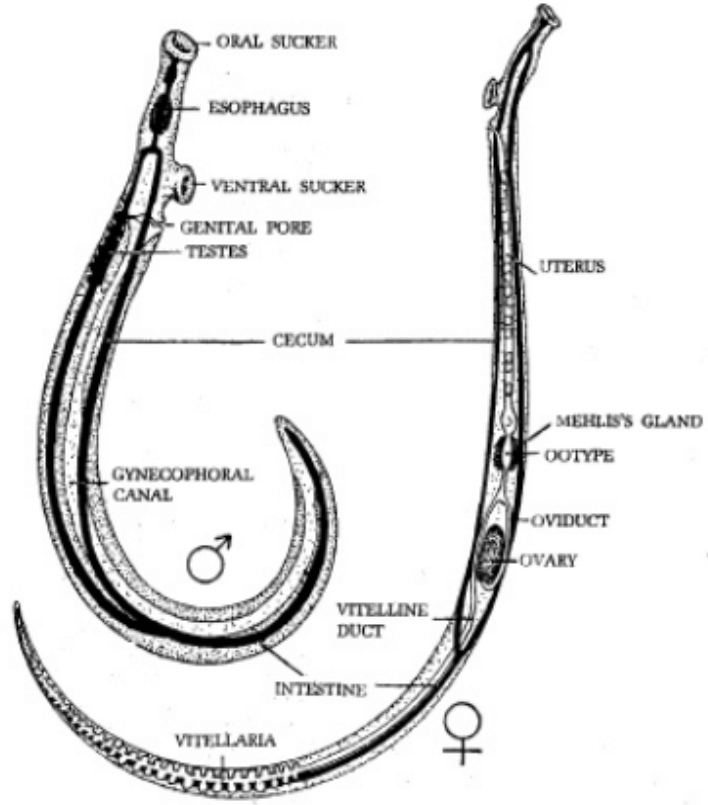
Male worms are shorter and stouter than females.

- males have

_____

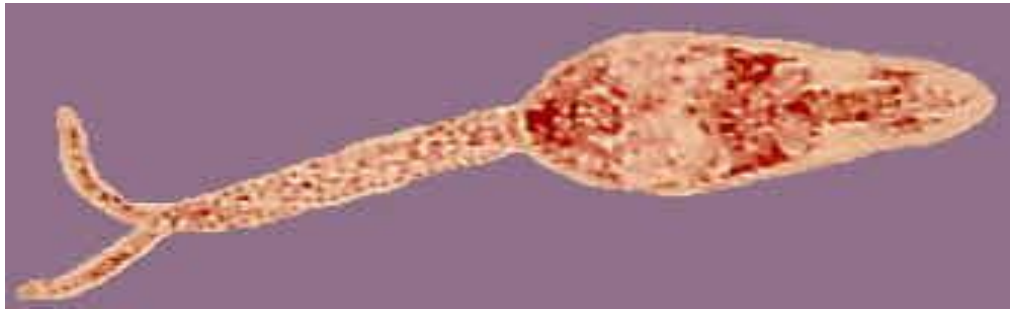
- ventral longitudinal groove in which the female resides

- **several testes** are located behind the acetabulum

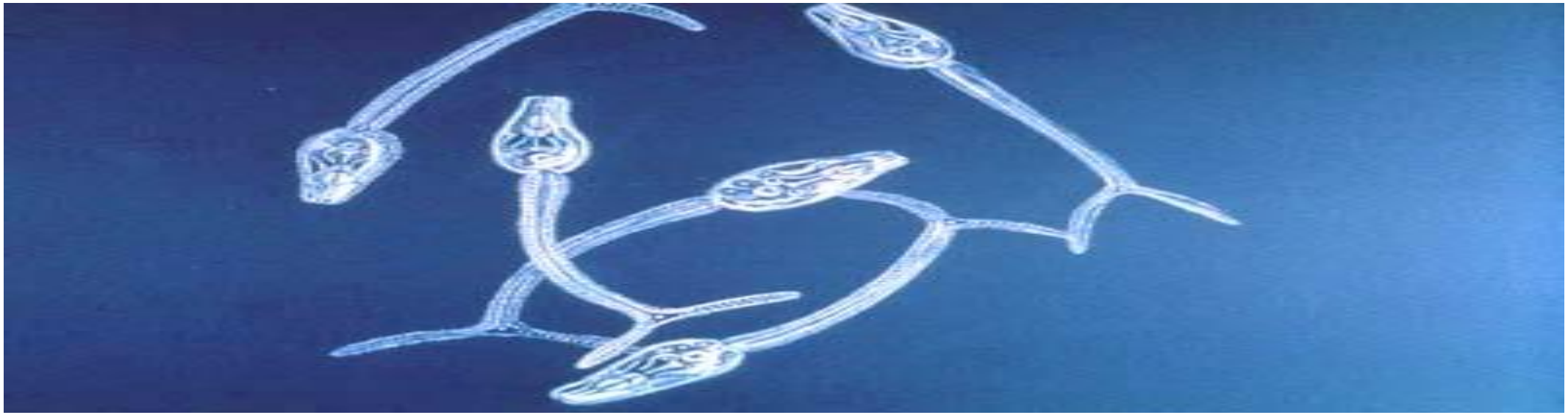


Cercaria larva of *Schistosoma spp.*

The infective stage for human and other final hosts are cercariae by penetration through the unbroken skin.





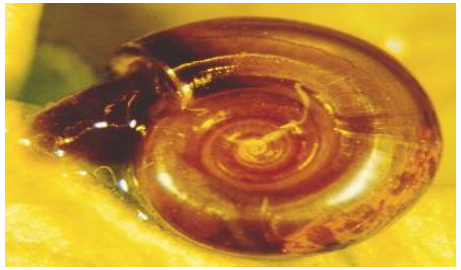


Life Cycle

- *Schistosoma spp.* passes its life cycle in 2 hosts.
- Definitive host: Humans are definitive hosts.
- Intermediate host: Fresh water snails.

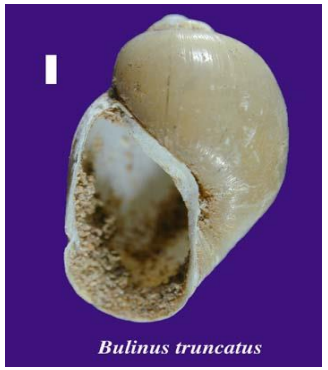
S. mansoni

(Biomphalaria)



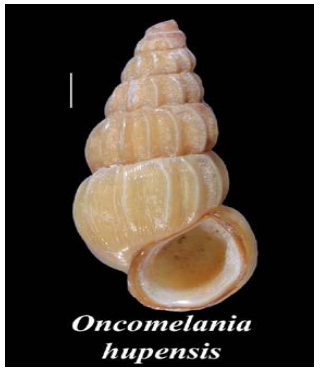
S. hematobium

(Bulinus)



S. japonicum

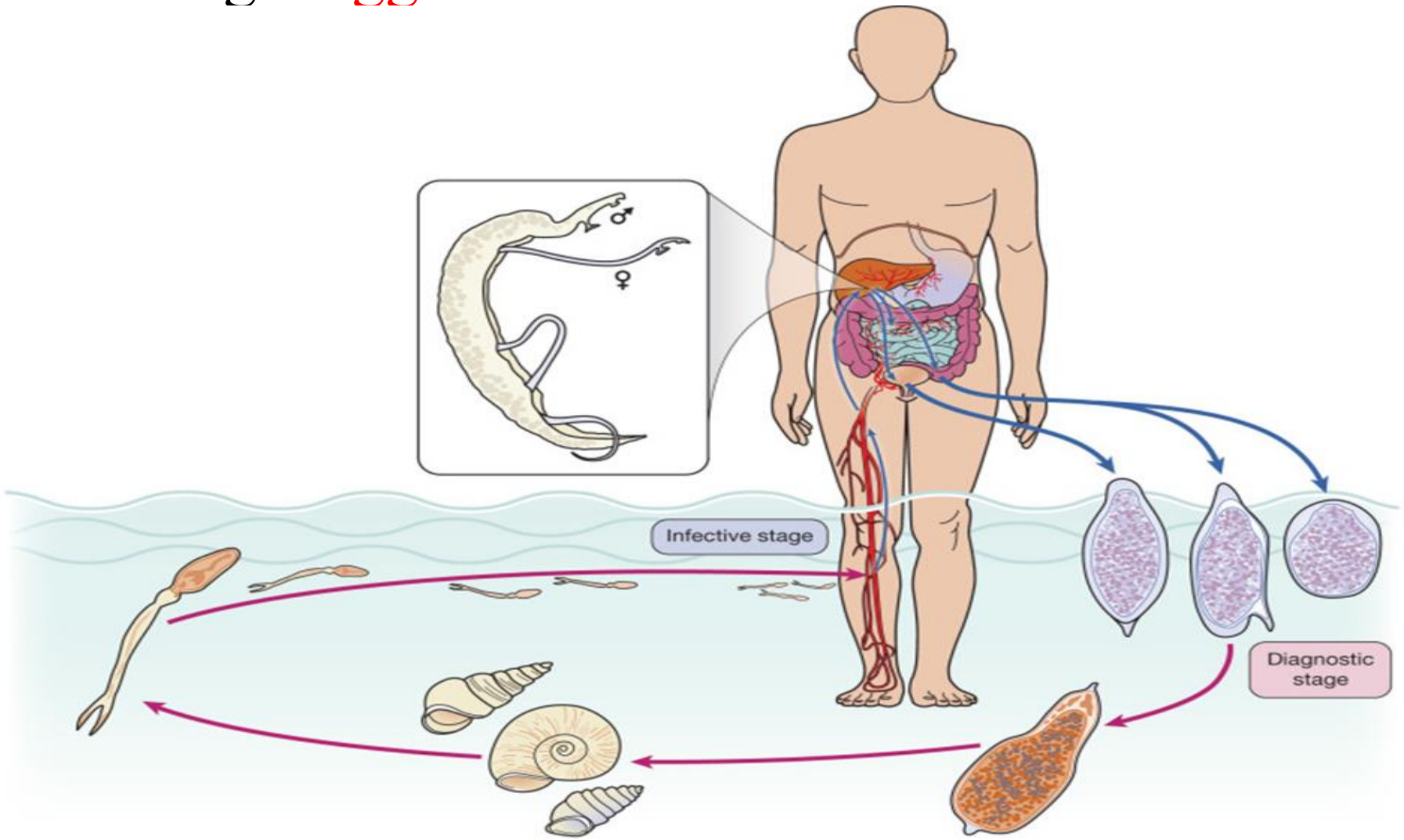
(Oncomelania)



- Infective form: Cercaria larva



Infective stage: cercariae
Diagnostic stage: eggs



Morphology of the eggs of the schistosomes that infect humans



***Schistosoma mansoni* egg**
Ellipsoidal with lateral spine

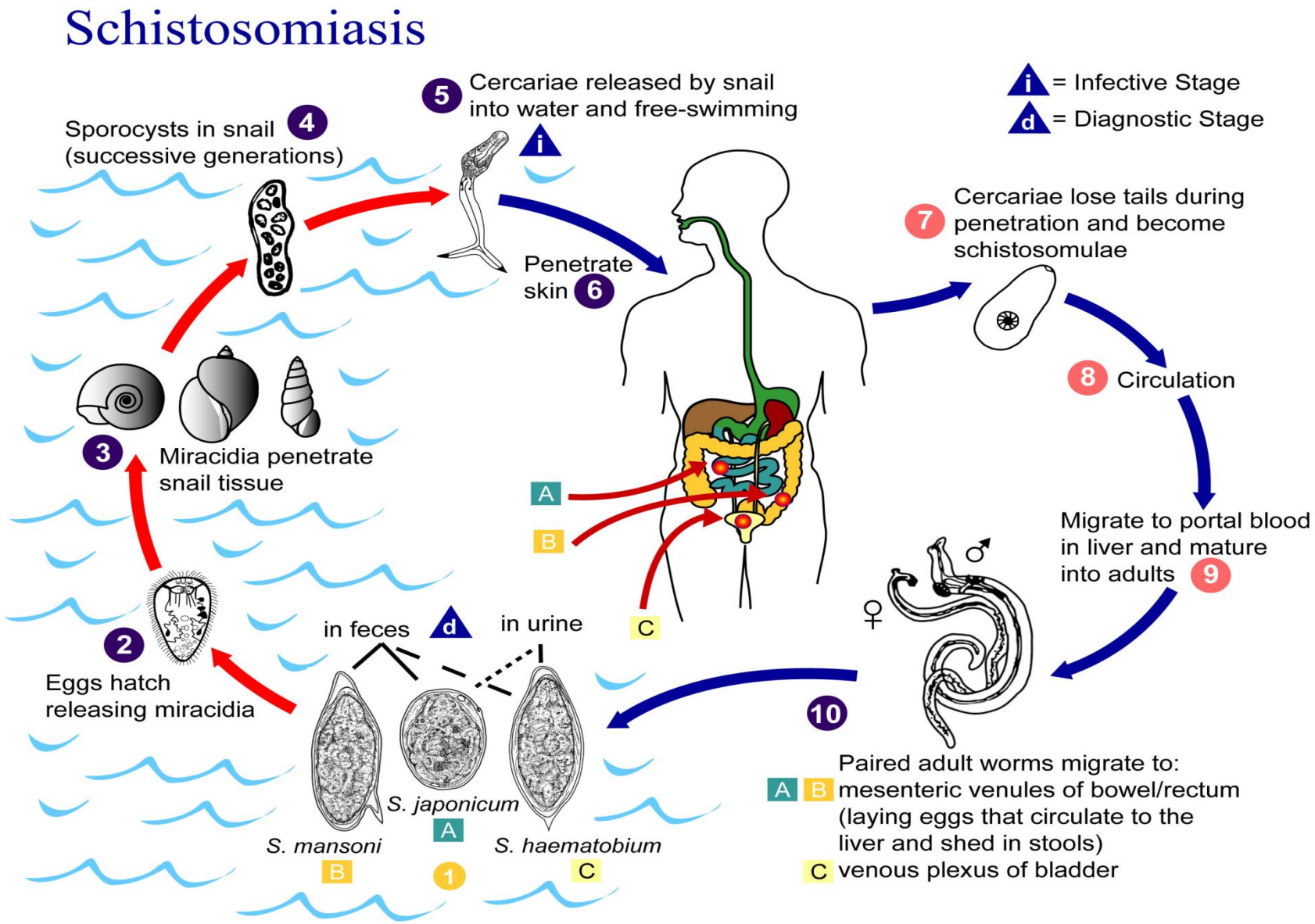


***Schistosoma haematobium* egg**
Ellipsoidal with terminal spine



***Schistosoma japonicum* egg**
Round with rudimentary spine

Life Cycle


Schistosomiasis
i = Infective Stage
d = Diagnostic Stage
5 Cercariae released by snail into water and free-swimming
Sporocysts in snail 4 (successive generations)
Penetrate skin 6
7 Cercariae lose tails during penetration and become schistosomulae
8 Circulation
Migrate to portal blood in liver and mature into adults 9
3 Miracidia penetrate snail tissue
A
B
C
in feces
in urine
2 Eggs hatch releasing miracidia
10
Paired adult worms migrate to:
A B mesenteric venules of bowel/rectum (laying eggs that circulate to the liver and shed in stools)
C venous plexus of bladder
S. japonicum
A
S. mansoni
B
S. haematobium
C
1

Pathogenicity and Clinical Features

- Clinical illness caused by schistosomes can be classified depending on the stages in the evolution of the infection, as follows:
- Skin penetration and incubation period.
- Egg deposition.
- Tissue proliferation.
- The clinical features during the incubation period (Due to the cercaria and schistosomulum) may be local cercarial dermatitis or general anaphylactic or toxic symptoms. Cercarial dermatitis consists of transient itching and petechial lesions at the site of entry of the cercariae (swimmer's itch).

The acute stage, which begins shortly after cercarial penetration (itching and dermatitis) followed 2 to 3 weeks later by fever, chills, diarrhea, and urticaria. lymphadenopathy, and hepatosplenomegaly. This condition can also called (Katayama fever).





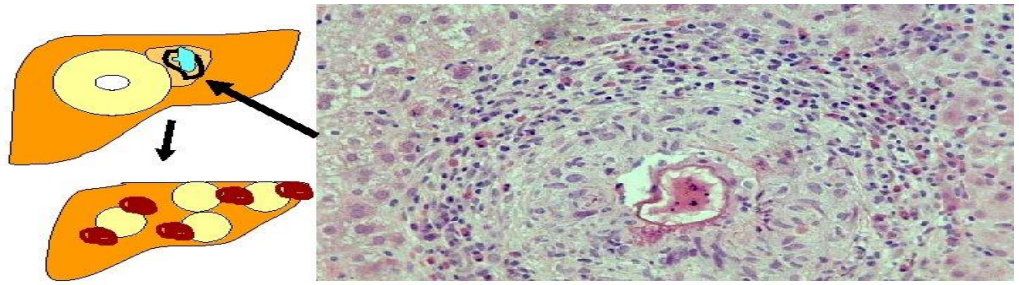


(swimmer’s itch).

Clinical features during oviposition (Egg deposition)

Chronic stage:

- Most of the serious pathologic findings are caused by the presence of eggs in the liver, spleen, or wall of the gut or bladder. Eggs in the liver induce granulomas, which lead to fibrosis, hepatomegaly, and portal hypertension.
- The granulomas are formed in response to antigens secreted by the eggs. Portal hypertension leads to splenomegaly.



Schistosomiasis: Pre-sinusoidal, intrahepatic portal hypertension

H &E stain







abdomen distention looks like a big drum, Ascites, and splenomegaly.

Clinical features during tissue proliferation

- In the chronic stage of genitourinary schistosomiasis, there is generalized hyperplasia and fibrosis of the vesical mucosa with a granular appearance (sandy patch).
- The eggs of *S. haematobium* in the wall of the bladder induce granulomas and fibrosis, which can lead to carcinoma of the bladder. (this can present with a symptoms such as blood in urine).

Cystoscopy of common bilharzial lesions in the urinary bladder.

(A) Bilharzial pseudotubercles and adjacent ulcer;

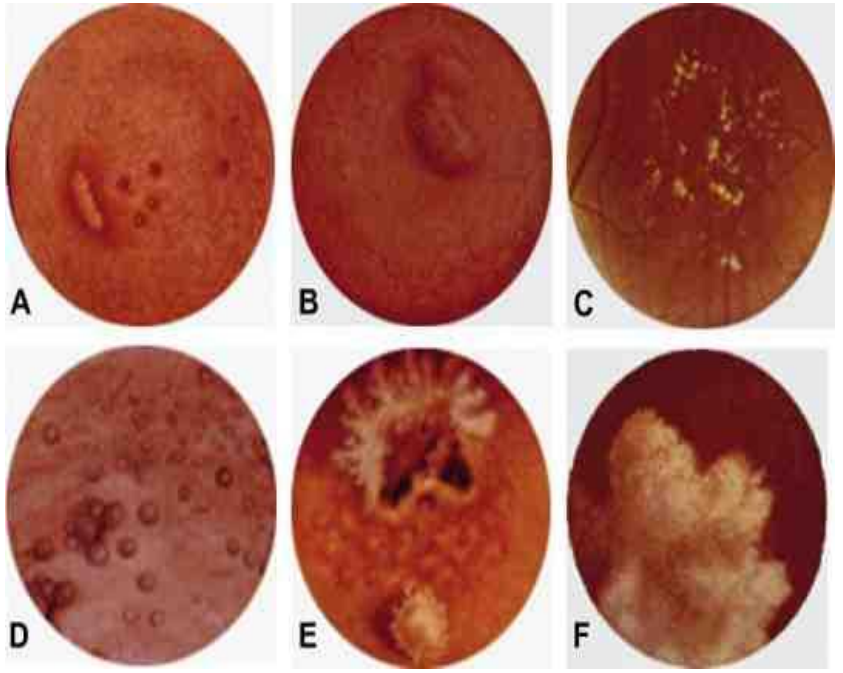
(B) Bilharzial sessile mass covered by psudotubercles

(C) Sandy patches

(D) cystitis cystica

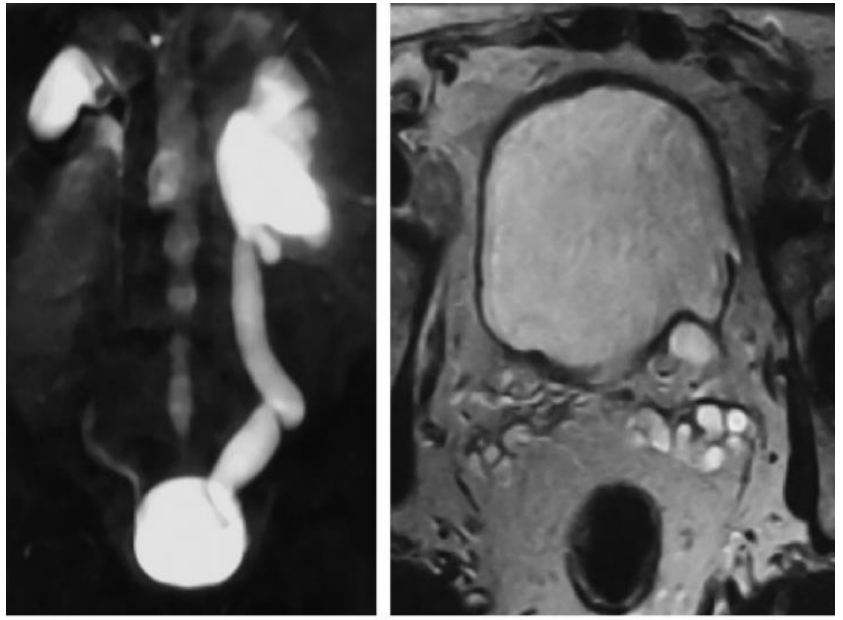
(E) malignant ulcer (squamous cell carcinoma) with adjacent phosphate encrustations and sandy patches.

(F) fungating malignant mass (squamous cell carcinoma)



Cystoscopy appearances of common bilharzial lesions in the urinary bladder

- There may be obstructive hyperplasia of the ureters and urethra (hydroureter).
- Chronic infection specially in children can lead to anemia.
- Secondary bacterial infection leads to chronic cystitis.



USG may show hydroureter and hydronephrosis.

- Calculi form in the bladder due to deposition of oxalate and uric acid crystals around the eggs and blood clots.
- significant disease may occur in other organs during schistosomiasis (Lungs and central nervous system (spinal cord). Genital organs may be involved).

Diagnosis of Schistosomes

Diagnosis depends on finding the characteristic ova in the feces or urine under the microscope

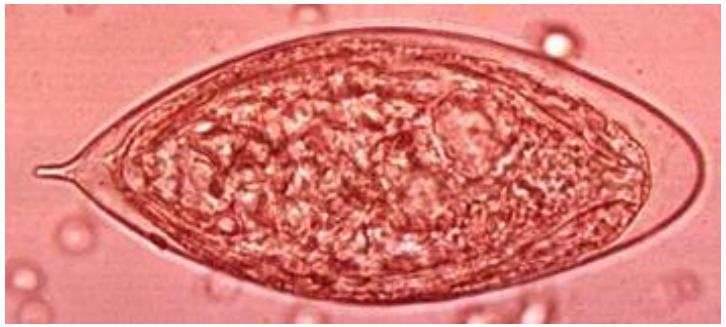
- ***Schistosoma mansoni***

  lateral spine



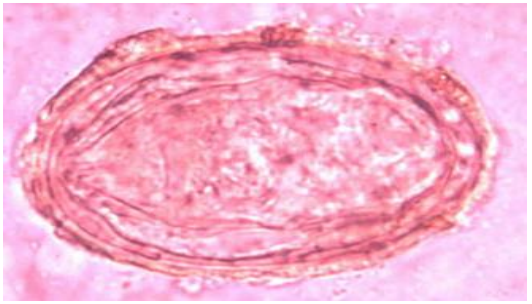
- ***Schistosoma haematobium***

  terminal spine



- ***Schistosoma japonicum***

  no spine

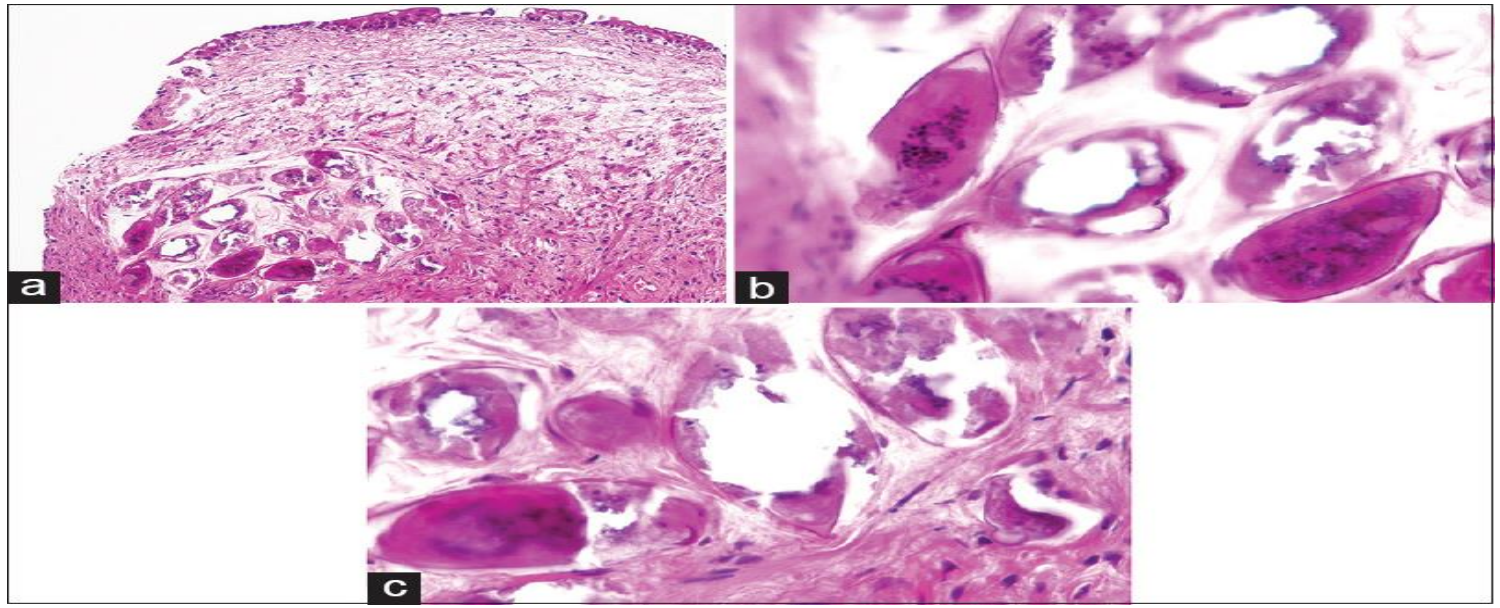


Histopathological examination:

Schistosome infection may also be diagnosed by demonstrating its eggs in

Bladder mucosal biopsy (*Schistosoma haematobium in bladder,*)-

Rectal mucosal biopsy (*Schistosoma mansoni* and *Schistosoma japonicum)*



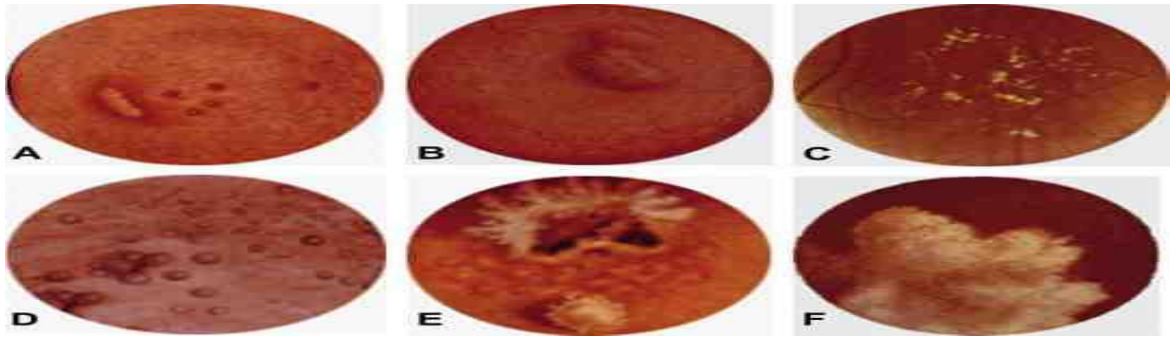
bladder mucosal biopsy shows

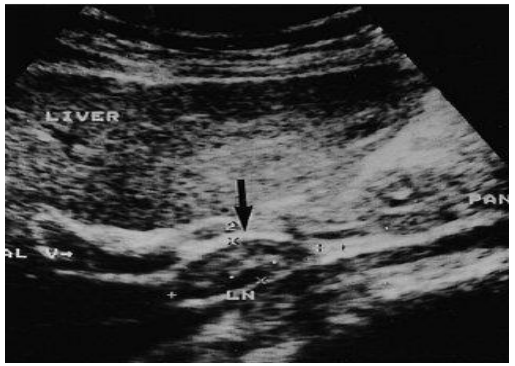
(a) clusters of parasitic eggs in the lamina propria of bladder mucosa
(b) The nuclei of miracidia
(c) characteristic terminal spines (center).

Serological Diagnosis: detection of specific schistosome antigens.

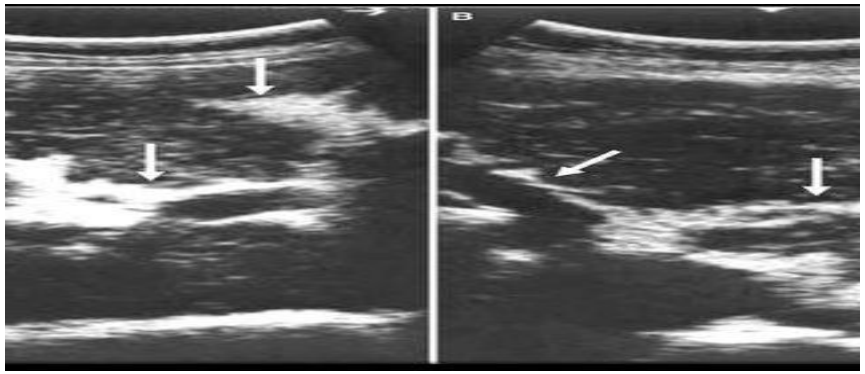
- Imaging techniques: (Ultrasound, CT, X-ray and MRI) are useful in the diagnosis of the Schistosomiasis.
- Cystoscopy are also useful in diagnosis of the disease.

Cystoscopy appearances of common bilharzial lesions in the urinary bladder.

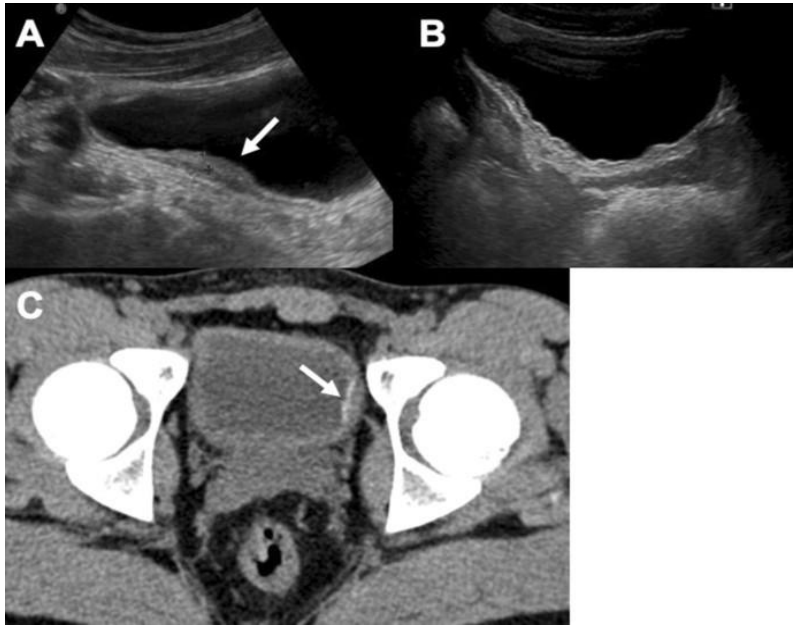




Pri-portal lymph node in patient with schistosomiasis



Urogenital schistosomiasis



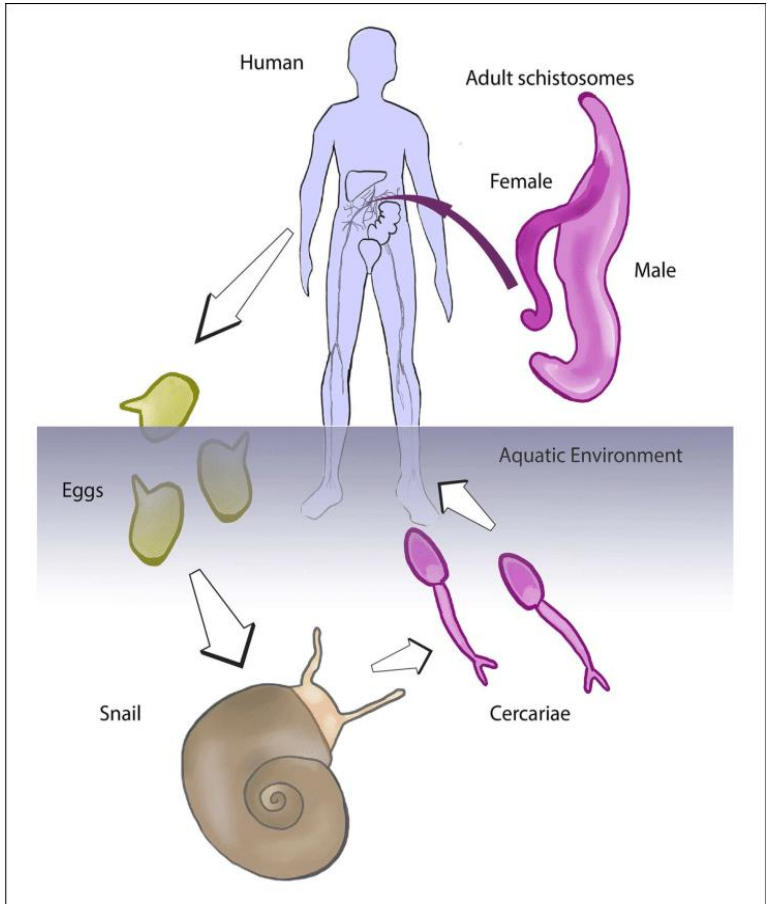
Ultrasound of schistosomiasis infected 40 years man

Treatment

- Praziquantel is the treatment of choice for all three species.

Prevention

- Prophylactic measures include:
- Eradication of the intermediate hosts (water snails).
- Prevention of environmental pollution with urine and feces.
- Effective treatment of infected persons.
- Avoid swimming, bathing, and washing in infected water.



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